age points using just 24 training images. On concrete aggregate grading, it improves from 12.0% (base model) to 79.6% with only 18 labeled images. Training completes in under 4 hours per task on 2×V100 GPUs, and under 2 hours on a single L4 GPU, making same-day adaptation feasible.

These results demonstrate that answer-conditioned CoT distillation is a practical method for manufacturing environments where inspection requirements change frequently and labeled data is scarce. By separating the frontier model's description ability from its classification accuracy, we generate correct training reasoning even for tasks where the teacher itself performs poorly. This enables same-day deployment of domain-adapted VLMs without large datasets or extended training cycles.

Future work includes ablation studies on the number of CoT descriptions per image, evaluation across additional VLM architectures and scales, and statistical significance testing with multiple seeds.

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